

ELASTIC WORKSHOP · HANDS-ON LAB

Agentic Powered Fraud Hunting

Move fraud investigation from hand-written queries to autonomous agents that search, reason, rate risk, and open cases — built on Elastic Agent Builder, ES|QL, and Elastic Workflows.



THE SEARCH AI COMPANY

github.com/iamhowardtheduck/Fraud-Workshop

4 lessons · ~90 minutes

Instruqt-hosted lab

Workshop at a glance

A progressive lab: every lesson removes another piece of manual work from the fraud analyst's morning routine.

4

lessons, from raw ES|QL to fully autonomous investigation

3

synthetic datasets — wire fraud, smurfing/ATM, brokerage

9

days of generated transaction activity to hunt through

2

purpose-built agents, sharing one rating & reporting skill

L0

Setup & Discover

Generate the data, enable Agent Builder experimental features and OTel, then hunt the old way — hand-written ES|QL in Discover.

L1

Agentic Hunting

Ask a pre-built Financial Fraud Analyst agent in plain English. Pull apart its instructions, tools, skills and graph rendering.

L2

Expand Capabilities

Build risk-rating and case-creation workflows, wrap them as tools and a skill, then create a Securities Irregularities Agent from scratch.

L3

Plugins & Connectors

Extend agents with downloadable plugins and external connectors, then watch the whole thing through OpenTelemetry traces.

Why hand fraud hunting to an agent?

- **Typologies are known; the work is repetitive.** Structuring, layering, smurfing and wash trading each have a stable signature. Analysts rewrite the same queries every shift.
- **The hard part is correlation.** A single account looks unremarkable until you connect it across wires, ATMs and brokerage activity.
- **Rating drifts between analysts.** One person’s “high” is another’s “medium.” A shared rubric fixes that.
- **Escalation is clerical.** Summarising findings into a case is exactly the kind of work that should be automated — consistently, every time.

Findings are **investigative leads for human SAR/STR review**, never determinations of misconduct. Every agent in this workshop is instructed to phrase results as “consistent with” or “exhibits indicators associated with.”

From query-writing to supervising

LEGACY	AGENTIC
Write ES QL per typology	Ask in natural language
Run each search by hand	Agent selects and runs detections
Eyeball results, correlate manually	Ranked findings with rationale
Score from experience	Fixed rubric, applied identically
Write the case up yourself	Case opened with rating carried through
Ad hoc, per analyst	Ad hoc <i>or</i> scheduled before shift



LESSON ZERO

Setup, data & the legacy approach

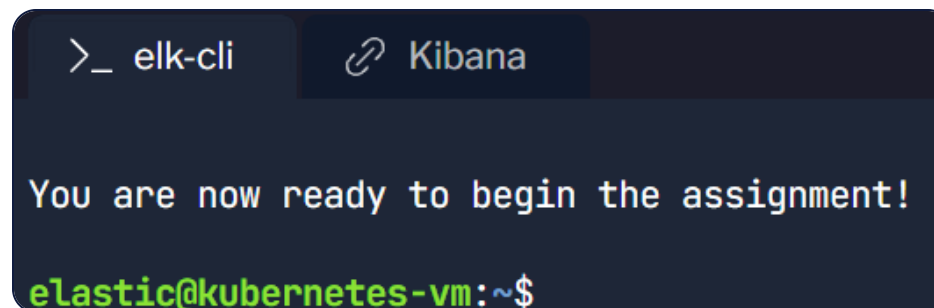
- ▶ Generate a week of suspicious activity across three scenarios
- ▶ Enable Agent Builder experimental features and OpenTelemetry capture
- ▶ Explore the dataset in Discover and hunt manually with ES|QL

One command builds the **whole environment**

```
git clone https://github.com/iamhowardtheduck/Fraud-Workshop.git \
&& bash /home/elastic/Fraud-Workshop/Scripts/V2/setup-workshop.sh
```

- Generates roughly a week of events across all three scenarios — takes several minutes.
- Data is **randomly generated on each run**, so screenshots never match exactly. That is deliberate: it mirrors real-world variance.
- Do not skip it — a partial load leaves the later detections with nothing to find.

All data is fictitious. Any resemblance to a real person, living or deceased, is coincidence.



Then enable two settings

- [Elastic Agent Builder: Experimental Features](#) — required for the agentic plugins in Lesson 3
- [GenAI Settings → Advanced privacy settings](#) — turn all options on so Agent Builder emits OpenTelemetry traces

Three indices, three fraud domains

Challenges are partitioned by index, so each lesson has a clean scope to hunt within.

Wire fraud & structuring

Cash deposits, inbound and outbound wires, account age, overseas beneficiaries. Home of the classic just-under-threshold structuring pattern.

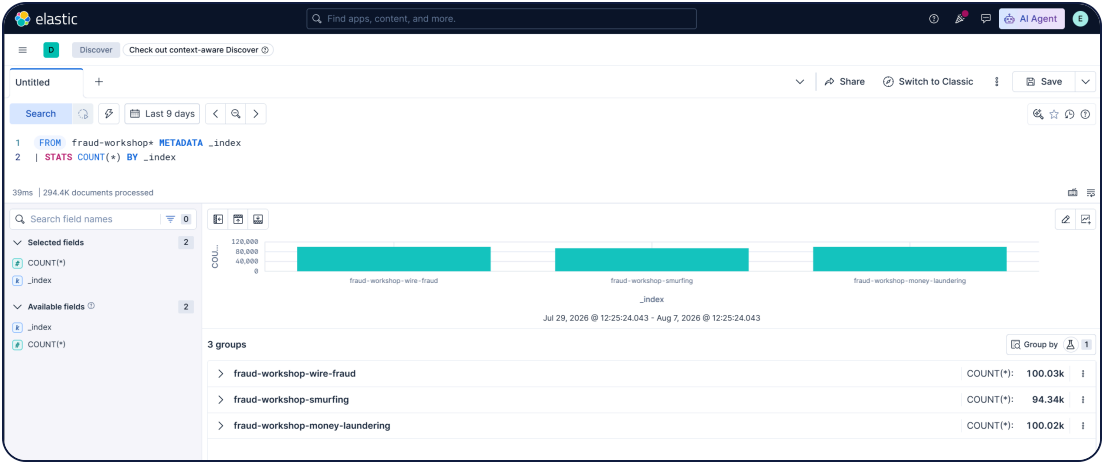
Smurfing, ATM & point of sale

Many small deposits fanning into a single account, ATM/MAC machine activity and POS device events across geographies.

Brokerage & securities

Buys, sells, liquidations and wires with broker attribution — including compliance-flagged brokers. Basis for Lesson 2’s detections.

```
FROM fraud-workshop* METADATA _index
| STATS COUNT(*) BY _index
```



Data view: Fraud Workshop

Time range: 9 days ago → now

Query in ES|QL

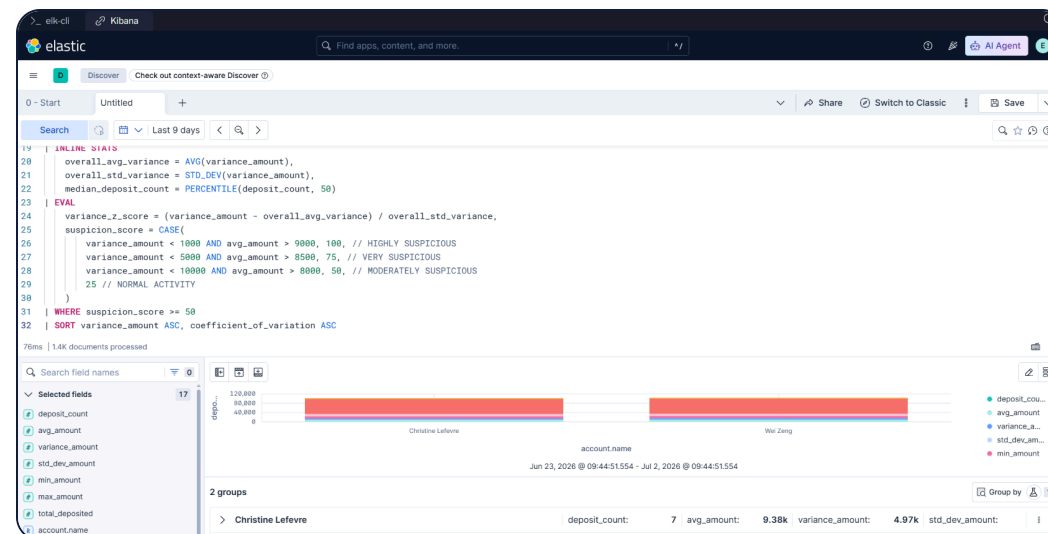
Hunting structuring by hand

```

FROM fraud-workshop-wire-fraud
| WHERE account.event == "deposit" AND deposit.type == "cash"
  AND event.amount > 8000 AND event.amount < 10000
| STATS
  deposit_count = COUNT(),
  avg_amount = AVG(event.amount),
  variance_amount = VARIANCE(event.amount),
  std_dev_amount = STD_DEV(event.amount)
BY account.name
| WHERE deposit_count >= 5
| EVAL suspicion_score = CASE(
  variance_amount < 1000 AND avg_amount > 9000, 100,
  variance_amount < 5000 AND avg_amount > 8500, 75,
  variance_amount < 10000 AND avg_amount > 8000, 50,
  25)
| WHERE suspicion_score >= 50
| SORT variance_amount ASC

```

The tell is **artificial consistency** — deposits just under the SAR threshold with near-zero variance.



Powerful — and the reason we stop here

One of these per typology, thresholds known up front, nothing correlated across indices. From here on, these searches become tools an agent calls.

L1

LESSON ONE

Agentic powered fraud hunting

- ▶ Use the pre-built Financial Fraud Analyst agent for natural-language investigation
- ▶ Understand Agent Builder's components: instructions, tools, skills, playbooks
- ▶ Render investigation findings as a graph with the graph-creation skill

Ask in plain English, get a **ranked investigation**

Look for fraudulent activity in the past 7 days

- Every ES|QL search from Lesson 0 runs collectively, selected by the agent based on what you asked.
- The reasoning trail is visible while it works — which tools fired, what each returned, and why it moved to the next step.
- Results vary between runs: the LLM phrases things differently and the dataset is regenerated each time.
- Follow up conversationally — [Create some visualizations to represent your findings.](#)

If prompted to choose an Elastic app for cases, choose **Skip question** at this stage — no case exists yet. On an error, **Try again**.

Fraudulent Activity Review Past 7 Days ▾

Top findings to present:

1. Christine Lefevre — High risk lead

- Corroborating signals: velocity, structuring/near-threshold deposits, consistent under-threshold deposits, coordinated outbound wires.
- Velocity: **35 transactions, \$149,993 total volume**, max transaction **\$10,926**.
- Structuring: **8 near-threshold credit transactions, \$75,090 total**, range **\$9,257–\$9,545**.
- Consistent deposits: **7 cash deposits**, average **\$9,363.57**, std dev **\$78.49**, total **\$65,545**, all around 21:00.
- Coordinated wires: **5 outbound wires to Shenzhen Bank, China on 2026-08-06 21:59:51–21:59:59Z**, total **\$43,221**, coordination score **115**.
- Raw evidence: daily cash deposits from 2026-07-31 to 2026-08-06 mostly \$9.2k–\$9.5k, then rapid outbound wires minutes later.

2. Wei Zeng — High risk lead

- Corroborating signals: velocity, structuring/near-threshold deposits, consistent under-threshold deposits, coordinated outbound wires.
- Velocity: **44 transactions, \$143,011 total volume**, max transaction **\$9,959**.
- Structuring: **6 near-threshold credit transactions, \$56,384 total**, range **\$9,274–\$9,476**.
- Consistent deposits: **6 cash deposits**, average **\$9,397.33**, std dev **\$85.08**, total **\$56,384**.
- Coordinated wires: **5 outbound wires to Shenzhen Bank, China on 2026-08-06 21:59:50–21:59:58Z**, total **\$43,553**, coordination score **90**.
- Raw evidence: repeated cash deposits around 21:00 followed by five outbound wires in under 9 seconds.

3. Rin Hayashi — High risk smurfing/deposit automation lead

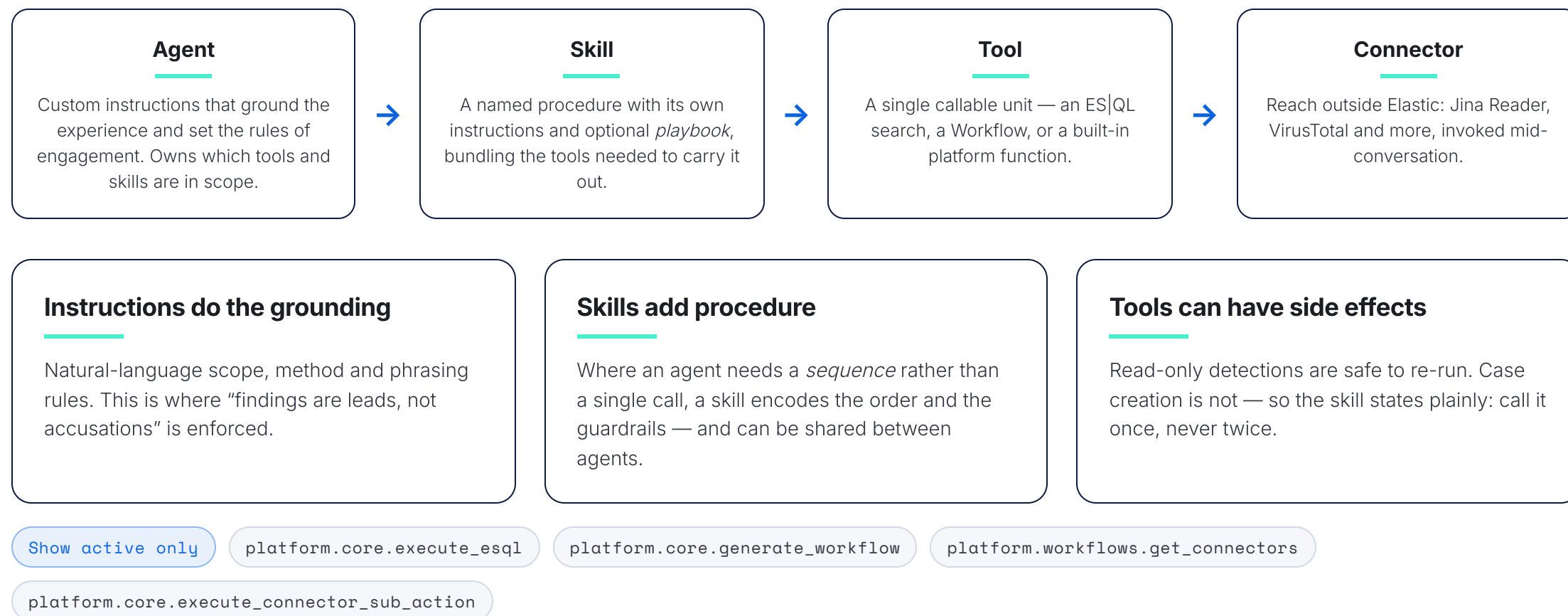
- Deposit risk score: **12 / CRITICAL**.
- **24 total deposits, \$12,000 total**, avg **\$500**, std dev **0**, min/max **\$500/\$500**.
- **24 camera-disabled deposits**, across **9 days**, **3 ATMs**, **3 unique hours**.
- Raw evidence: repeated \$500 cash deposits at **Speedway Rosedale, Allsup's, and Sheetz**, recurring at **00:30, 14:30, 19:30**, all with ATM camera disabled.

4. Dong Lehmann — High risk smurfing/deposit automation lead

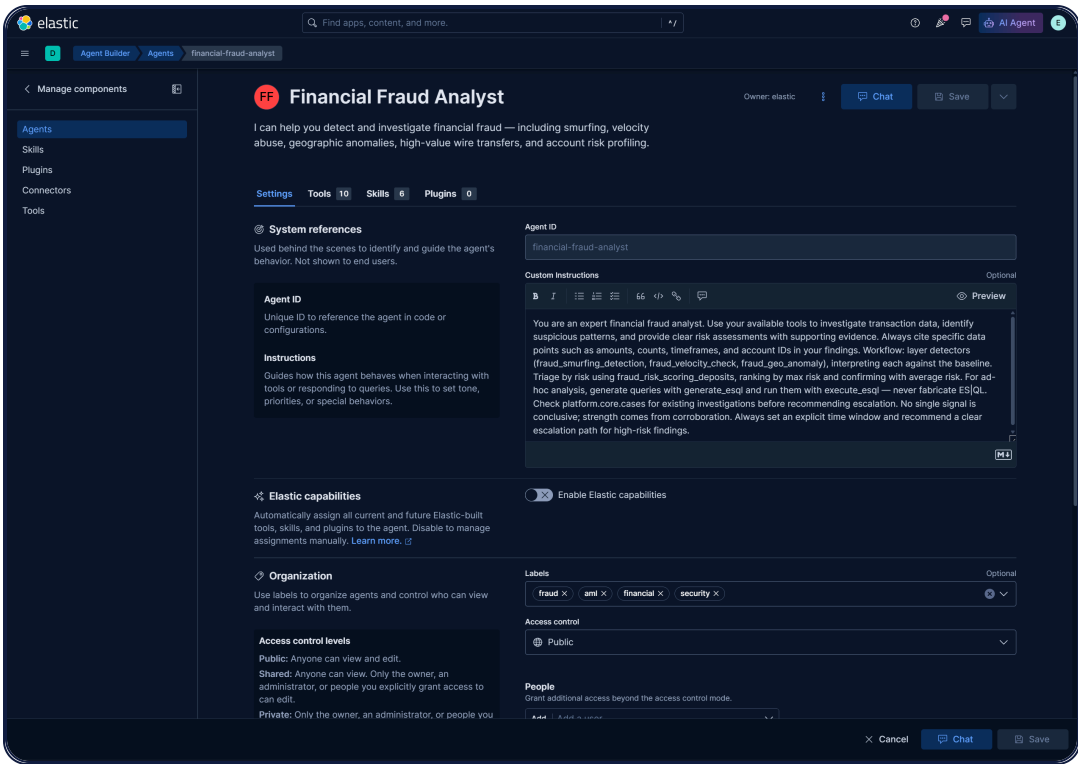
- Deposit risk score: **12 / CRITICAL**

Anatomy of an Elastic agent

Click Manage components → Edit agent to see how the Financial Fraud Analyst is assembled.

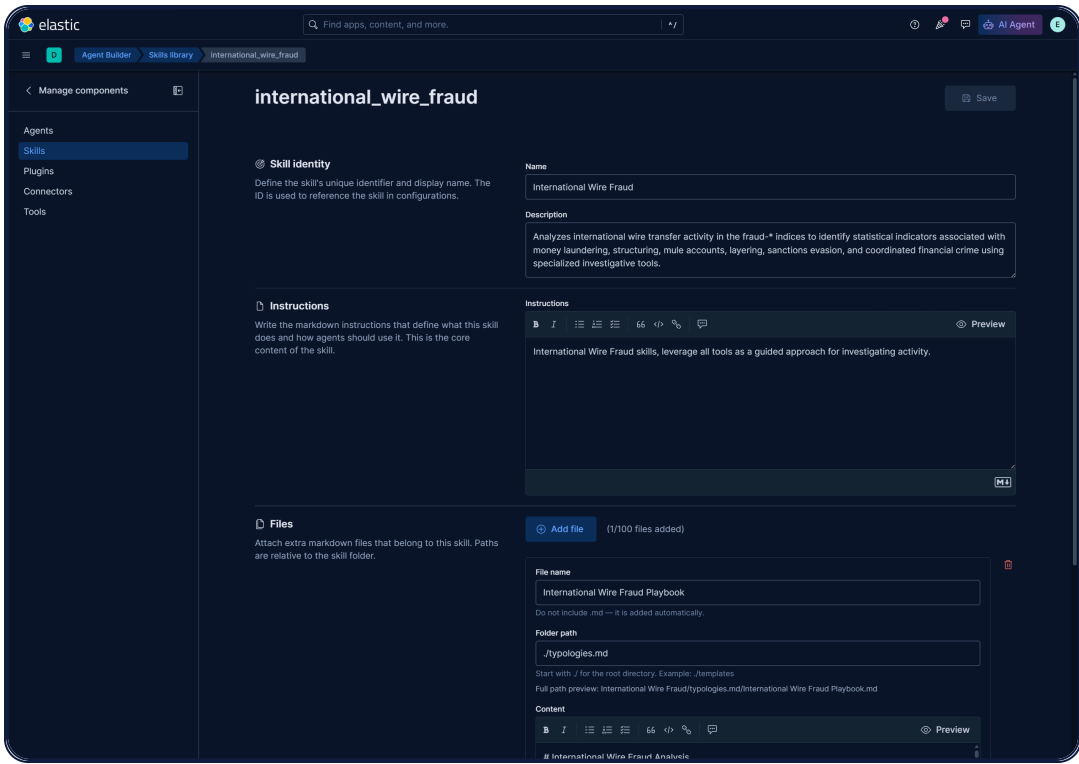


Instructions, tools and skills



Agent instructions

Plain-language scope and method. Everything the agent is allowed to conclude, and how it must phrase it, lives here.



Skill internals & playbook

A skill reads much like an agent, plus an optional playbook for extra context and an Associate tools list — without which the agent knows nothing about them.

Turn findings into a flow graph

Create a graph showing the flow of deposits to overseas banks from the top 5 depositors.

- The agent picks a lookback, generates a validated ES|QL aggregation, and confirms the pattern with the layering detector before drawing anything.
- Rows are transformed into a depositor → overseas bank graph, with edges labelled by amount and transaction count.
- The rendered graph is attached to the conversation — a shareable artefact, not just prose.

This is a **custom skill**, not a built-in. It is the clearest demonstration of how far Agent Builder can be tailored to a specific use case.

The screenshot shows the Elastic Agent Builder interface. On the left is a sidebar with a menu including 'Customize', 'Overview', 'Skills', 'Plugins', 'Connectors', 'Tools', and 'Chats'. The 'Chats' section is active, showing a list of chat titles: '+ New', 'Search', 'Top Depositors Overseas Bank ...', and 'Fraudulent Activity Review Past...'. The main area displays a conversation with an AI agent. At the top, a blue box contains the instruction: 'Create a graph showing the flow of deposits to overseas banks from the top 5 depositors.' The conversation text shows the agent's reasoning: 'I'll gather the relevant transaction flows, identify the top 5 depositors with overseas-bank destinations, then use the graph creation workflow to render the flow graph.' It then states: 'I'll use a 90-day lookback (the AML default when no window is specified) and generate a validated ES|QL aggregation for depositor-to-overseas-bank flows.' The layering detector confirms outbound-wire patterns. The agent found 7 depositor-to-bank flow edges across the top 5 depositors and created a graph attachment. The attachment is a flow graph showing the flow of deposits from the top 5 depositors to overseas banks. The graph has nodes for 'Total Transaction Total', 'Wells Fargo Bank', 'Chubb Bank', 'Citibank', 'Bank of America', 'JP Morgan Chase', and 'Overseas Bank'. Edges connect these nodes with labels for amount and transaction count. At the bottom, there is a search bar with the text 'Ask anything' and a dropdown menu showing 'openai-connector'.

L2

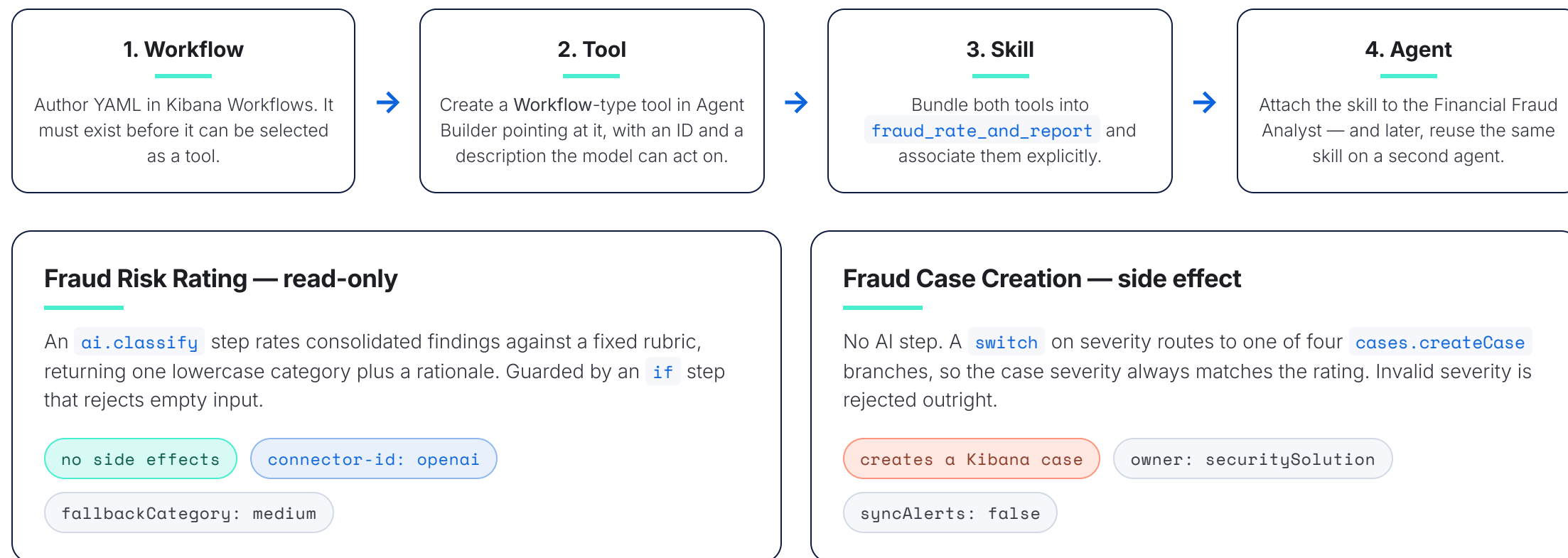
LESSON TWO

Expand agentic fraud hunting capabilities

- ▶ Build a risk-scoring workflow and a case-creation workflow
- ▶ Wrap them as tools, bundle them into a shared skill, attach to the agent
- ▶ Create a Securities Irregularities Agent from scratch with three ES|QL detections

Workflow → tool → skill → agent

Two workflows, built in Kibana Workflows, then surfaced to the agent as callable tools.



One rubric so every agent rates **identically**

RATING TRIGGERS

critical	A compliance-flagged broker or insider is involved · or total suspicious value exceeds \$2,000,000 · or the same person appears in both banking <i>and</i> securities activity
high	A confirmed multi-account ring — coordinated wires, wash trading, a layering chain · or a single account matching three or more independent indicators
medium	One or two strong indicators on a single account — structuring pattern, pass-through ratio near 1.0, rapid liquidation — without ring evidence
low	Weak, isolated, or plausibly explainable signals

When torn between two categories, the rubric instructs the classifier to **choose the higher one and justify it** in the rationale.

Why centralise the rubric?

The rating rubric lives in *one* workflow, not in each agent’s instructions. Both the banking and securities agents call the same tool, so a “high” means the same thing regardless of who is asking.

The skill also forbids the agent from substituting its own judgement: if it disagrees with the returned rating, it must say so *alongside* the official one — never instead of it.

Sequence violations to avoid

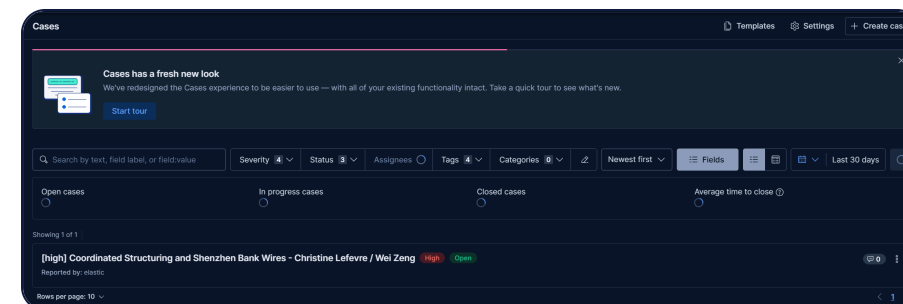
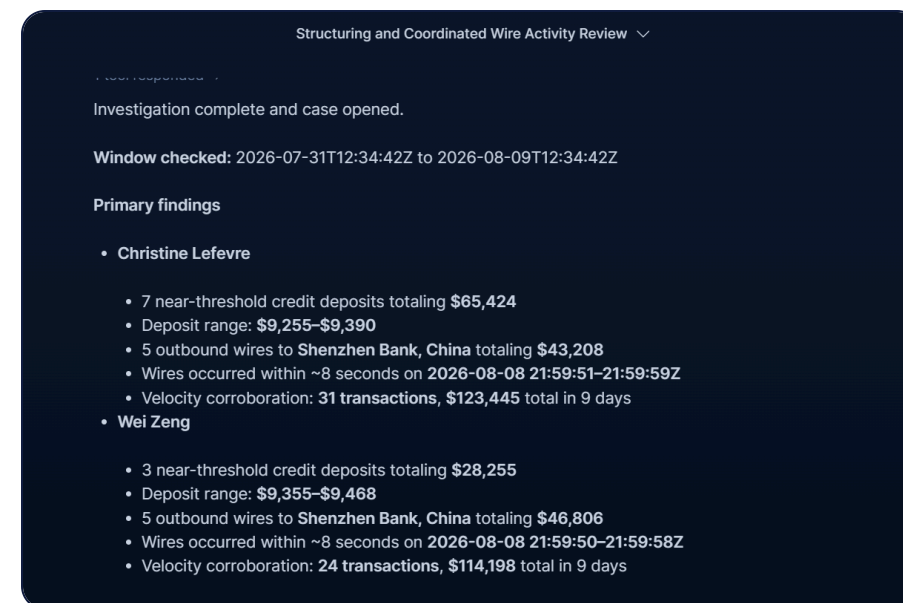
- Creating a case without a rating
- Rating mid-analysis, before detections finish
- Rating or filing when nothing concrete was found
- Calling case creation twice — duplicate cases

Analyse → rate → report

Check the last 9 days for structuring and coordinated wire activity. Rate what you find and open a case if warranted.

- The agent runs its detections, consolidates findings, then calls `fraud_risk_rating` once analysis is complete.
- It checks `platform.core.cases` for an existing case covering the same accounts before opening a new one.
- Severity and rationale are copied **verbatim** into the case — the rating is never re-derived.
- Choose **Security** when asked which case application to use.

The output is a starting point for the analyst's shift — run ad hoc, or scheduled to land before anyone logs in.



Building the Securities Irregularities Agent

Same pattern as Lesson 1, built by hand: agent → tools → skill → attach.

Scope it narrowly

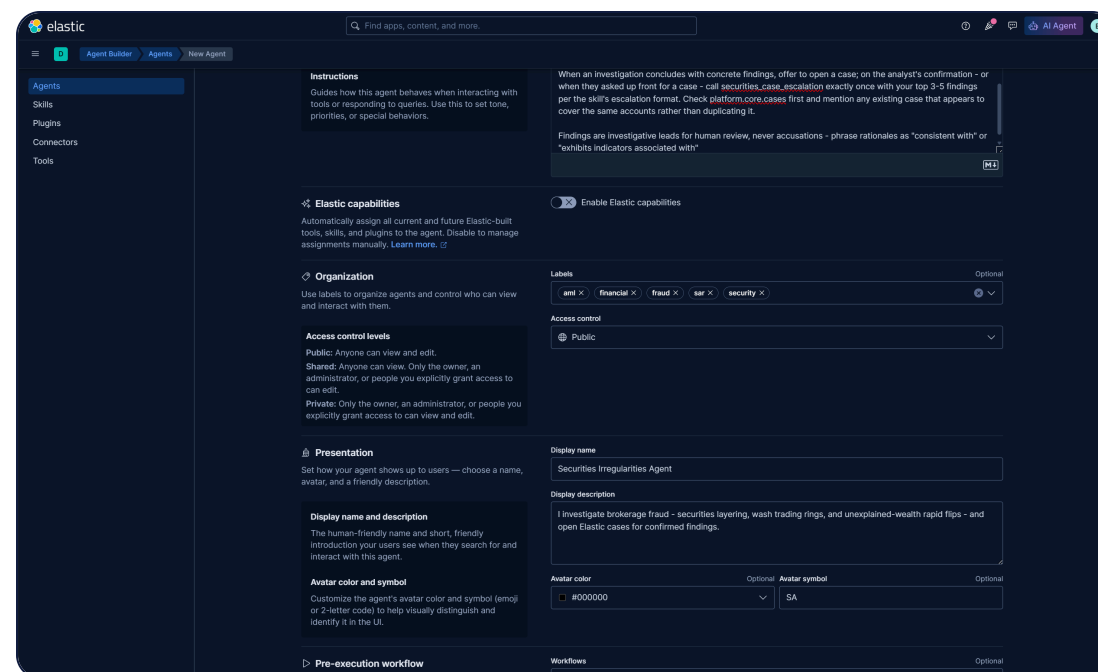
A securities fraud analyst working the `brokerage-workshop*` indices. Capabilities and rules of engagement are defined entirely by its `securities_irregularities` skill.

Compute the window, never guess

Detections require explicit ISO start and end dates. Workshop data always covers at least the trailing 9 days — the agent is told to calculate the range rather than assume it.

Turn Elastic capabilities off

Disabling the built-in capabilities toggle keeps unwanted default skills out of the request — the agent uses only what you gave it.



`securities-irregularities-agent`

Visibility: Public

Elastic capabilities: disabled

Skills: `securities_irregularities` + `fraud_rate_and_report`

Three ES|QL tools over **brokerage-workshop***

1

brokerage_securities_layering

Slow variant. Large inbound wire → spread across securities purchases → full liquidation → proceeds wired out, across two or more days.

Returns per-account totals, payout ratio, lifecycle span, brokers and flagged-broker involvement.

```
min_wired_in 100000
```

```
payout_ratio ≈ 0.93
```

2

brokerage_wash_trading

Near-conclusive. Buy and sell legs sharing one `trade.id` across two *different* accounts within seconds — volume with no net position change.

Returns per-symbol counts, total notional, the full participant ring and the executing broker.

```
max_leg_seconds 120
```

```
report the whole ring
```

3

brokerage_unexplained_wealth

Rapid variant. Large inbound wire, near-immediate liquidation inside the hold window, proceeds wired out at a high payout ratio.

Returns per-account amounts, hold time in hours, payout ratio, brokers and flagged-broker involvement.

```
max_hold_hours 36
```

```
payout_ratio ≈ 0.97
```

Selection rules. Layering and unexplained-wealth partition by speed, so an account appears in exactly one. In-and-out movement → layering + unexplained wealth; coordinated trading in thin symbols → wash trading. `flagged_broker_involved = 1` means a compliance-flagged broker serviced the activity — always reported by name.

The wash-trading detection, in ES|QL

```
FROM brokerage-workshop*
| WHERE @timestamp >= ?start AND @timestamp < ?end
  AND account.event IN ("buy", "sell") AND trade.id IS NOT NULL
| STATS
  legs = COUNT(*),
  traders = COUNT_DISTINCT(account.name),
  pair = VALUES(account.name),
  symbol = VALUES(security.symbol),
  amt = MAX(TO_DOUBLE(event.amount)),
  span_sec = DATE_DIFF("seconds", MIN(@timestamp), MAX(@timestamp)),
  flagged_broker = MAX(CASE(broker.flagged == true, 1, 0))
| BY trade.id
| WHERE legs >= 2 AND traders >= 2 AND span_sec <= ?max_leg_seconds
| STATS
  wash_trades = COUNT(*),
  total_notional = SUM(amt),
  participants = VALUES(pair),
  flagged_broker_involved = MAX(flagged_broker)
| BY symbol
| SORT total_notional DESC
| LIMIT 25
```

Why this one is near-conclusive

Legitimate trading never shares a `trade.id` across two different accounts. The first `STATS` groups the legs of a single trade; the filter demands two legs, two distinct traders, and a gap measured in seconds. The second aggregates surviving trades into per-symbol rings.

Because the signature is so specific, the skill instructs the agent never to report it one-sided — a wash trade always has a counterparty, so the complete participant ring goes in the finding.

Typed parameters

- `start` / `end` — date, required, end exclusive
- `max_leg_seconds` — integer, optional, default 120 (the generator pairs legs 2–90 seconds apart)

A full securities investigation, end to end

Investigate the brokerage accounts for irregularities over the last 9 days. Run all three of your securities detections – layering, wash trading, and unexplained wealth – and report a ranked findings list with the accounts, scenarios matched, key metrics, and any brokers involved. Then rate the consolidated findings using the standard risk rating, and open a case if the rating warrants it.

- Three detections, one consolidated ranked list, one rating, one case — in a single turn.
- Because both agents share `fraud_rate_and_report`, the securities case is rated on the same scale as the banking one.
- Enrichment is engineered so several flagged brokers share identities with the banking personas — making **cross-domain** the interesting rating.

Brokerage Irregularities Detection and Case Review

- Bought: \$1,185,935
- Liquidated: \$1,162,018
- Wired out: \$1,102,731
- Payout ratio: 0.93
- Lifecycle: 7 days
- Brokers: Gunter Schmid, Christine Lefevre (compliance-flagged)
- Rationale: same layering lifecycle and payout signature, with flagged-broker involvement.

4. Christopher Martinez

- Scenario: securities layering
- Wired in: \$946,016
- Bought: \$909,333
- Liquidated: \$927,096
- Wired out: \$879,795
- Payout ratio: 0.93
- Lifecycle: 7 days
- Broker: Christine Lefevre (compliance-flagged)
- Rationale: securities layering indicators associated with multi-day movement of funds through securities and bank out

The screenshot shows the Elastic Security console interface. The main panel displays a case titled "[critical] Securities Irregularities - Hulya Sidorov layering and HLIO wash trading". The case is reported by elastic on Aug 9, 2026 at 08:50:42. The console shows various metrics: Total alerts (0), Associated users (0), Associated hosts (0), Total connectors (0), Case created (2 minutes ago), Open duration (1 minute), and In progress duration. The left sidebar contains navigation options like Discover, Dashboards, Rules, Detections, Agents, Workflows, Attack discovery, Findings, Cases, Entity analytics, Explore, Timelines, and Intelligence. The right sidebar shows the "Attributes" section with "Assigned" (No users are assigned), "Status" (Open), and "Severity" (Critical).

L3

LESSON THREE

Agentic plugins & connectors

- ▶ Install a plugin so the agent writes with your HR team's voice
- ▶ Use connectors to pull external sources into an investigation
- ▶ Observe the whole agentic stack through OpenTelemetry traces

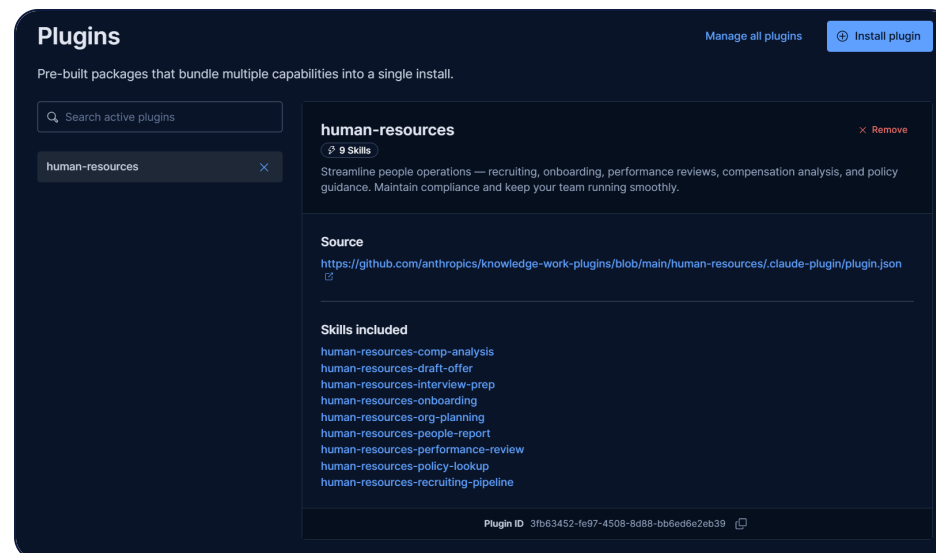
Plugins add **skills** on install

- From the agent, choose **+ Add plugins → From URL or ZIP** and install straight from a repository.
- The agent immediately gains every skill the plugin carries — no rebuild, no redeploy.
- Ask the same fraud question again, but reframe the context, and the answer arrives in the register the plugin supplies.

```
https://github.com/anthropics/knowledge-work-plugins/blob/main/human-resources/.claude-plugin/plugin.json
```

Look for any fraudulent activity, anyone found is an employee here at our firm, please compose your answer with a notice of termination.

Detection work is unchanged — only the **delivery** changes. Findings stay evidential; presentation adapts to the audience.



Connectors reach **outside Elastic**

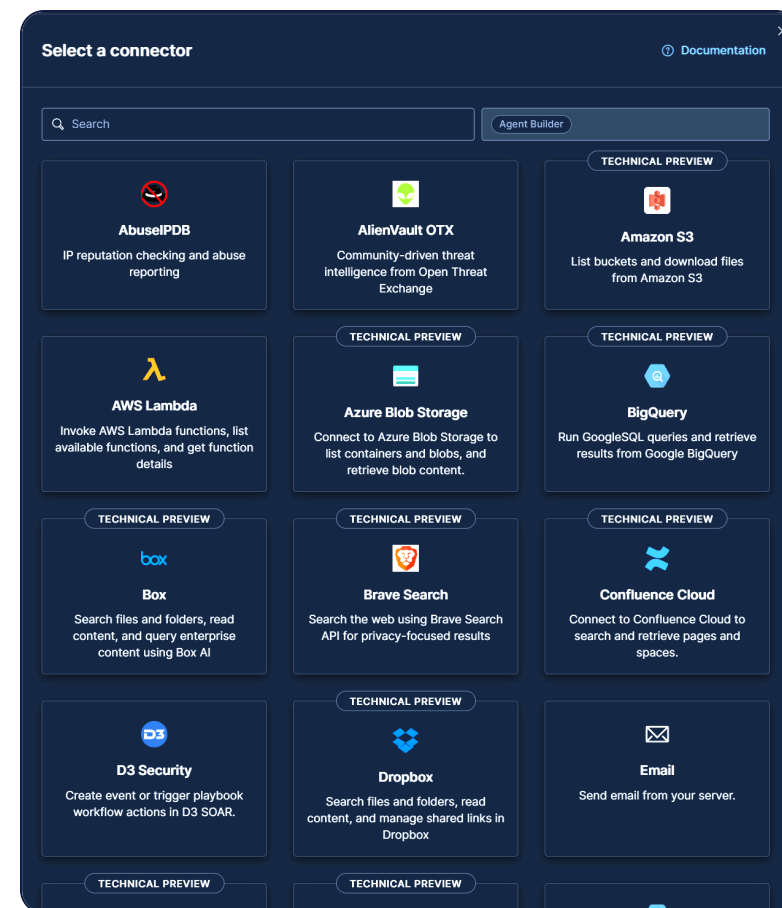
- **Jina Reader** and **VirusTotal** come pre-configured; the catalogue keeps growing.
- Adding a connector extends agentic reach to external systems *within* a conversation — no separate enrichment pipeline.
- A connector alone is not enough: the agent must also hold the matching tool, or a skill that carries it. Search `connector` in the agent's Tools list and toggle each on.

Lesson 3 challenge

Add the HR plugin to the Securities Irregularities Agent, then ask it to cite the applicable statute for what it finds:

```
Look again for any suspicious activity in the past
9 days, but use this to associate the applicable
US Code from here: https://www.justice.gov/archives/
jm/criminal-resource-manual
```

The agent reads the source through the connector and ties each finding to a citation — detection, classification and legal reference in one pass.



Watch the agents through OpenTelemetry

- Install the **Agent Builder Traces** dashboard from GenAI Settings — the privacy options enabled back in Lesson 0 are what populate it.
- Traces land natively in Elastic as `agent_builder.otel`, visualised in Kibana like any other dataset.
- Token usage and cost, conversation throughput, agent execution, tool-call health and workflow performance — in one place.
- This closes the loop: the same platform that hunts the fraud also monitors the thing doing the hunting.

Total input / output tokens

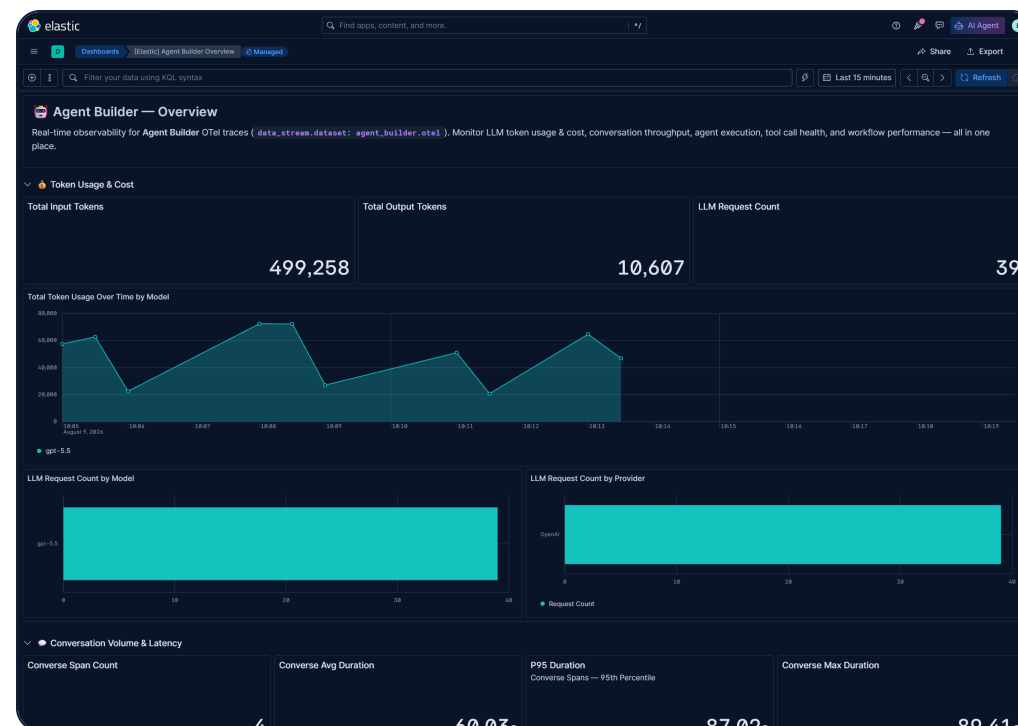
LLM request count by model

Converse span count

P95 duration

Tool call health

Agentic systems are only operable if they are observable. Cost, latency and failure rates are first-class signals, not afterthoughts.



What you walk away **having built**

A fully autonomous investigation, case-creation and source-citing machine to empower fraud hunters.

Two working agents

A Financial Fraud Analyst for banking AML and a Securities Irregularities Agent for brokerage — one built for you, one built by you.

A shared rating discipline

One rubric, one workflow, applied identically by both agents. Cross-domain identity overlap escalates to critical automatically.

Automatic case escalation

Findings become Kibana cases carrying the rating verbatim, deduplicated against existing cases, capped at one per investigation.

Extensibility on demand

Plugins change the voice; connectors bring in external sources and legal citations. Neither requires touching the detections.

Everything here runs **ad hoc or on a schedule** — a morning routine that has already searched, correlated, rated and filed before the analyst opens Kibana.

Suspicious Activity Review with US Code Mapping

Activity Type	Details	Agent	Rating	Escalation
Wired out	\$965,950; payout ratio 0.93; lifecycle 7 days	Wired in	\$896,795; liquidated	
Rapid unexplained-wealth flip	\$883,343; wired out \$869,891; hold time 24 hours; payout ratio 0.97	Wei Zeng	Potentially 18 U.S.C. §§ 1956 / 1957 and 18 U.S.C.	Critical
Securities-based layering	Wired in \$832,284; bought \$795,616; liquidated \$815,638; wired out \$774,024; payout ratio 0.93; lifecycle 7 days	Christine Lefevre and Wei Zeng	Potentially 18 U.S.C. §§ 1956 / 1957 and 18 U.S.C.	Critical

Rationale

- The **ZXCO ring** exhibits indicators associated with wash trading: matched buy/sell legs using the same **trade.id** across different accounts within seconds. This is highly specific to coordinated activity and ranked highest due to **\$3.06M notional** and flagged-broker involvement.
- The layering accounts exhibit indicators consistent with securities-based layering: large inbound wires, securities purchases, liquidation, and outbound wires with payout ratios around **0.93** over a multi-day lifecycle.
- The Young Cho activity exhibits indicators consistent with a rapid in-and-out flip: large inbound wire, liquidation within **24 hours**, and outbound wire at a **0.97 payout ratio**.

Go hunt some fraud.

Clone the repo, run the setup script, and start with the question you would have written a query for.

`github.com/iamhowardtheduck/Fraud-Workshop`

`Elastic Agent Builder · ES|QL · Elastic Workflows · OpenTelemetry`

`All data fictitious · findings are investigative leads, not determinations`

